

t2g square lattice tight-binding derivation

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1 Single Layer

We look at hopping from all orbitals along the two x-y directions.

1.1 xy band

1.1.1 x-direction

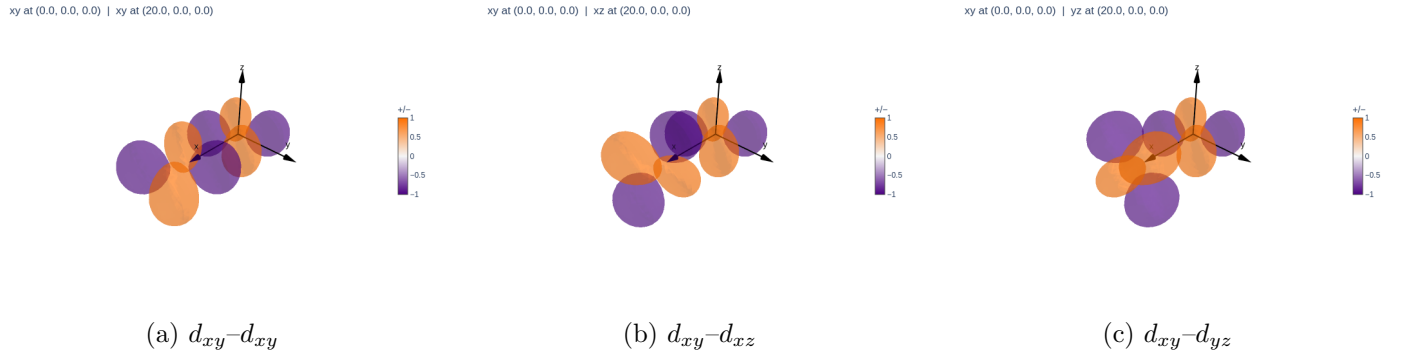


Figure 1: t2g orbital overlaps for nearest-neighbour hopping along x .

We look at the slater-koster integrals. The slater direction cosines are

$$l = \frac{\mathbf{R} \cdot \hat{x}}{|\mathbf{R}|} \quad m = \frac{\mathbf{R} \cdot \hat{y}}{|\mathbf{R}|} \quad n = \frac{\mathbf{R} \cdot \hat{z}}{|\mathbf{R}|} \quad (1)$$

$$(2)$$

In the x direction this is just.

$$l = 1 \quad m = 0 \quad n = 0 \quad (3)$$

This simplifies the xy-xy, xy-xz, and xy-yz integrals to

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \quad (4)$$

1.1.2 y-direction

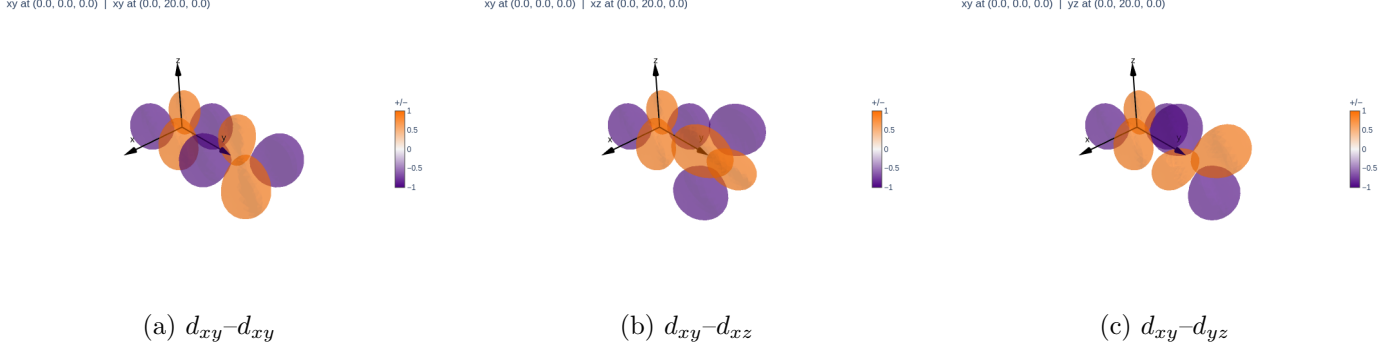


Figure 2: t2g orbital overlaps for nearest-neighbour hopping along y .

For the y -direction the slater direction cosines are

$$l = 0 \quad m = 1 \quad n = 0 \quad (5)$$

and the integrals are

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \quad (6)$$

1.1.3 Tight Binding Hamiltonian

Putting this all together we get this hopping term in the hamiltonian and apply the Fourier transformation

$$c_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_i} c_{\mathbf{k}}$$

$$\begin{aligned} H &= t \sum_{\mathbf{R}, \delta} c_{i+\delta}^\dagger c_i \\ &= \frac{t}{N} \sum_{\mathbf{R}, \delta} \sum_{\mathbf{q}, \mathbf{k}} e^{-i\mathbf{q} \cdot (\mathbf{R}_i + \delta)} e^{i\mathbf{k} \cdot \mathbf{R}_i} c_{\mathbf{q}}^\dagger c_{\mathbf{k}} \\ &= \frac{t}{N} \sum_{\delta} \sum_{\mathbf{q}, \mathbf{k}} \delta_{\mathbf{q}\mathbf{k}} e^{-i\mathbf{q} \cdot \delta} c_{\mathbf{q}}^\dagger c_{\mathbf{k}} \\ &= t \sum_{\delta} \sum_{\mathbf{k}} e^{-i\mathbf{q} \cdot \delta} c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \\ &= t \sum_{\mathbf{k}} \left[e^{-ik_x a} + e^{-ik_y a} + e^{ik_x a} + e^{ik_y a} \right] c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \\ &= 2V_{dd\pi} \sum_{\mathbf{k}} [\cos(k_x a) + \cos(k_y a)] c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \end{aligned} \quad (7)$$

1.1.4 Next Nearest Neighbor Hopping

Now lets consider this one for the xy band. The hopping vector is now $(a\hat{x} + a\hat{y})$. Our directional cosines are

$$l = 1/\sqrt{2} \quad m = 1/\sqrt{2} \quad n = 0 \quad (8)$$

and the non zero integrals are

$$t_{xy-xy} = \frac{3}{4} V'_{dd\sigma} + \frac{1}{4} V'_{dd\delta} \quad (9)$$

where the primes denote NNN integrals. It turns out this is the result for all directions i.e $(\pm a\hat{x} + \pm a\hat{y})$ our tight binding hamiltonian can then be written as.

$$\begin{aligned}
H &= t_{nnn} \sum_{\delta} \sum_k e^{-ik \cdot \delta} c_k^{\dagger} c_k \\
&= t_{nnn} \sum_k \left[e^{-ia(k_x + k_y)} + e^{-ia(k_x - k_y)} + e^{-ia(-k_x + k_y)} + e^{-ia(-k_x - k_y)} \right] c_k^{\dagger} c_k \\
&= t_{nnn} \sum_k [2 \cos(k_x a + k_y a) + 2 \cos(k_x a - k_y a)] c_k^{\dagger} c_k \\
&= 4t_{nnn} \sum_k \cos(k_x a) \cos(k_y a) c_k^{\dagger} c_k
\end{aligned} \tag{10}$$

Using the sum to product $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$

1.1.5 Energy for Band

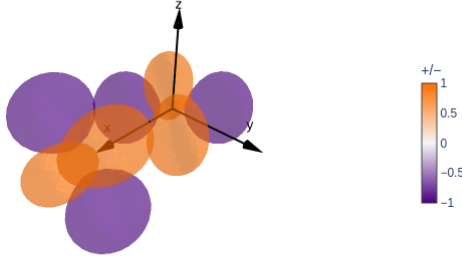
Hence the total energy for the xy band is

$$\varepsilon_{xy} = -2t_1 [\cos(k_x a) + \cos(k_y a)] - 4t_2 \cos(k_x a) \cos(k_y a) \tag{11}$$

1.2 yz band

To look at the yz band we apply the following permutation. Its helpful to look at this diagram

xy at (0.0, 0.0, 0.0) | yz at (20.0, 0.0, 0.0)



From xy we get yz by permuting the axis like $x \rightarrow y \rightarrow z \rightarrow x$, which means our direction cosines become $l \rightarrow m \rightarrow n \rightarrow l$

1.2.1 x direction

$$l = 0 \quad m = 1 \quad n = 0 \tag{12}$$

$$t_{yz,yz} = V_{dd\pi} \quad t_{yz,xy} = 0 \quad t_{yz,xz} = 0 \tag{13}$$

The hamiltonian is

$$\begin{aligned}
H &= t \sum_{\delta} \sum_k e^{-ik \cdot \delta} c_k^{\dagger} c_k \\
&= t \sum_k \left[e^{-ik_x a} + e^{ik_x a} \right] c_k^{\dagger} c_k \\
&= 2V_{dd\pi} \sum_k \cos(k_x a) c_k^{\dagger} c_k
\end{aligned} \tag{14}$$

1.2.2 y direction

$$l = 0 \quad m = 0 \quad n = 1 \tag{15}$$

$$t_{yz,yz} = V_{dd\delta} \quad t_{yz,xy} = 0 \quad t_{yz,xz} = 0 \tag{16}$$

The derivation for the energy is the same as above just in y direction.

1.2.3 Next Nearest Neighbor Hopping

We do the permutation to the directional cosines in the xy case to get

$$l = 0 \quad m = \pm 1/\sqrt{2} \quad n = \pm 1/\sqrt{2} \tag{17}$$

the diagonal hopping term is

$$t_{yz,yz} = \frac{1}{2} V'_{dd\pi} + \frac{1}{2} V'_{dd\delta} \tag{18}$$

which is the same for all diagonal directions so this gives us an additional diagonal term

$$\varepsilon_{yz,nnn} = -4t_{yz,yz} \cos(k_x a) \cos(k_y a) \tag{19}$$

Now the off diagonal gives us $t_{yz,xz} = 0$ but a nonzero

$$t_{yz,xy} = \pm \frac{1}{2} [V'_{dd\pi} - V'_{dd\delta}] \tag{20}$$

where it is plus/minus depending on the product of the direction cosines, i.e in our hopping hamiltonian we see

$$\begin{aligned}
H &= t_{yz,xy} \sum_{\delta} \sum_k e^{-ik \cdot \delta} c_k^{\dagger} c_k \\
&= t_{yz,xy} \sum_k \left[e^{-ia(k_x + k_y)} - e^{-ia(k_x - k_y)} - e^{-ia(-k_x + k_y)} + e^{-ia(-k_x - k_y)} \right] c_k^{\dagger} c_k \\
&= t_{yz,xy} \sum_k 2i [\sin(k_x + k_y) + \sin(k_x - k_y)] c_k^{\dagger} c_k \\
&= 4it_{yz,xy} \sum_k \sin(k_x a) \cos(k_y a) c_k^{\dagger} c_k
\end{aligned} \tag{21}$$

1.3 Energy for band

Hence the total energy is

$$\varepsilon_{yz} = -2t_1 \cos(k_x a) - 2t_{\delta} \cos(k_y a) \tag{22}$$

$$\langle yz | H_0 | xy \rangle = -4it_{yz,xy} \sin(k_x a) \cos(k_y a) \tag{23}$$

1.4 xz band

The same formalism applies.

1.5 x dir

$$l = 0 \quad m = 0 \quad n = 1 \quad (24)$$

which gives the nonzero integral $V_{dd\delta}$

1.6 y dir

$$l = 1 \quad m = 0 \quad n = 0 \quad (25)$$

which gives the nonzero integral $V_{dd\pi}$.

1.7 Next Nearest Neighbor Hopping

The directional cosines are

$$l = 1/\sqrt{2} \quad m = 0 \quad n = 1/\sqrt{2} \quad (26)$$

1.8 Energy For Band

$$\varepsilon_{xz} = -2t_\delta \cos(k_x a) - 2t_1 \cos(k_y a) \quad (27)$$

1.9 Summary

$$\begin{aligned} \varepsilon_{xy} &= -2t_1 [\cos(k_x a) + \cos(k_y a)] - 4t_2 \cos(k_x a) \cos(k_y a) \\ \varepsilon_{yz} &= -2t_1 \cos(k_x a) - 2t_\delta \cos(k_y a) \\ \varepsilon_{xz} &= -2t_\delta \cos(k_x a) - 2t_1 \cos(k_y a) \end{aligned}$$